

## Theme: The CAMPFIRE RESCUE

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# Theme 1

## [The Campfire Rescue]

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**By the end of the theme  
students will be able to:**

- ✓ List the main parts of a fire truck.
- ✓ Recall the basic functions of each fire truck component.
- ✓ Create 3D models of fire truck parts using Fusion 360 software.
- ✓ Apply mechanical design principles to develop accurate and functional prototypes.
- ✓ Synthesize knowledge and skills to construct a fully functional fire truck model.
- ✓ Describe how temperature sensors work and their role in detecting fires.
- ✓ Explain the basic principles of GPS technology and its application in emergency response.
- ✓ Analyse the interaction between temperature sensors, GPS modules, and the fire truck model.

# UNIT 01: [Beginner]

## Chapter 01: [Mechanics]

### Lesson 01: [The Campfire Rescue]

**Time Frame:** [90 mins]

#### Students will be able to:

- **Define** what fire is and describe the three elements of the fire triangle (heat, fuel, and oxygen).
- **Identify** and **recognize** different types of fire based on their sources (e.g., electrical, grease, paper, etc.).
- **Classify** fires into the correct fire classes (Class A, B, C, D, and K) and **explain** the appropriate extinguishing methods for each class.
- **Create** a poster that demonstrates knowledge of fire safety rules, precautions, and emergency response procedures.

- What do you think will happen to a fire if we remove one part of the fire triangle (heat, fuel, or oxygen)? Can you give an example of how this might happen in real life?
- A fire starts in the kitchen from cooking oil and spreads to an electric toaster. What types of fire are involved? What is the safest way to handle this situation?
- Why is using water on some types of fire, like oil or electrical fires, a bad idea? Which classes of fire should never be put out with water, and what should be used instead?

- **Self-Awareness**

*By understanding what fire is and how it starts, students become more aware of potential dangers around them and their own role in staying safe.*

- **Self-Management**

*Learning how to respond in a fire situation helps students manage their emotions (like fear or panic) and make safe, responsible decisions in emergencies.*

- **Social Awareness**

*Recognizing that fire safety isn't just personal—but also about protecting others—helps students develop empathy and a sense of responsibility for their family, classmates, and community.*

- **Relationship Skills**

*Group poster creation and discussions encourage teamwork, communication, and listening to others' ideas about safety.*

- **Responsible Decision-Making**

*Identifying the right way to handle different types of fires teaches students how to evaluate situations, consider safety outcomes, and choose the best course of action.*

## Lesson Plan

### 1. Engage

- **Teacher Role:**

#### Story

**Storyteller / Facilitator:**

Narrate the story with expression and suspense to spark curiosity.

**Ask Guided Questions:**

What do you think Adam should have known?

Why might it be important to know the type of fire?

What could have gone wrong if the wrong method was used?

**Connect to Learning Goals:**

Lead students to realize that knowing different fire types is a real-world skill.

- **Expected From Students:**
- **Active Listening:**  
Students listen carefully to the story.
- **Critical Thinking:**  
Respond to open-ended questions about what Adam could have done differently.
- **Sharing Ideas:**  
Students suggest what they already know about fire or fire safety.
- **Making Predictions:**  
Predict what might happen next in the story or how the fire could be controlled.

- **Intro Story:**

#### Strategies to Implement

- **Think-Pair-Share:**

After the story, students discuss with a partner:

- What kind of fire might it have been?
- What questions would you ask Adam if you were the fire operator?

- **KWL Chart (Know – Want to Know – Learned):**

Start with “K” and “W”:

- What do you know about fire?

- What do you want to know after hearing the story?

### • Role-Playing / Freeze Frame:

Have two students act out the phone call between Adam and the fire station.  
Freeze the scene and ask the class what should happen next.

### • Question Wall:

Create a space for students to post questions they have about fire types and safety.

## Story Reflection

1. What did Lila notice that was alarming?

- A) A bear.
- B) Smoke coming from the wood.
- C) A falling tree.
- D) A lost child.

2. Who did Adam call to report the fire?

- A) The police station.
- B) The fire station.
- C) The hospital.
- D) Their parents.

3. Why was Adam confused when talking to the dispatcher?

- A) He didn't know the location.
- B) He couldn't describe the type of fires.
- C) He forgot the phone numbers.
- D) He was too tired to talk.

### Reflection

1. What did Lila notice that was alarming?

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## Strategies to Implement

### 1. Think-Write-Share

- **How it works:**  
Students **think individually** about each question, **write their answers**, then **share** with a partner or small group.
- **Why it works:**  
Encourages both individual reflection and peer learning.

### 2. Individual Response Cards

- **How it works:**  
Students write their responses to the 3 questions on pre-prepared index cards or worksheets. The teacher collects them to assess understanding and emotional response.
  - **Why it works:**  
Encourages honest responses, especially from shy students. Supports formative assessment.
- 

### 3. Anonymous Digital Form (if tech is available)

- **How it works:**  
Use Google Forms, Padlet, or another tool to let students submit their answers digitally.
  - **Why it works:**  
Allows for privacy, real-time feedback analysis, and encourages full participation.
- 

### 4. Pair & Compare

- **How it works:**  
Students answer the 3 questions individually, then pair up to discuss their answers and note any differences in perspective.
  - **Why it works:**  
Builds communication and comparison skills while reinforcing learning.
- 

### 5. Reflect and Rate

- **How it works:**  
After answering the questions, students rate their understanding of the topic on a scale (e.g., 1–5 or smiley faces).  
Optional: Add a fourth question like "What do you still wonder?"
- **Why it works:**  
Helps students self-assess and gives the teacher insight into who may need further support.

## Think

“There are no right or wrong answers in this part, put here are some of the possible answers students can give”

**Do you think we use the same way to put out every type of fire?**

**Possible Answer:**

- No, each type of fire needs a different method.
- Using the wrong method, like water on an oil fire, can make it worse.
- That’s why it’s important to know the fire classes and how to respond to each.

**How can we extinguish these different types of fires?**

**Possible Answers:**

- Use a fire extinguisher that matches the type of fire (e.g., Class A, B, C, etc.).
- **Grease or oil fires (like the stove):** Never use water — use a fire blanket or Class B extinguisher.
- **Electrical fires:** Turn off the power if safe, and use a Class C fire extinguisher.
- **Wood or paper fires (like the forest or trash):** Use water or a Class A extinguisher.
- **Chemical or fuel fires:** Use a foam or dry chemical extinguisher.

**Looking at the image, what do you think you should do in these situations?**

**Possible Answers:**

- Stay calm and move away from the fire.
- Call the fire station immediately.
- Do not try to put out the fire unless I know it is safe and I have the right tools.
- Warn others nearby and get to a safe place.



Strategies to  
Implement



## 1. Think-Pair-Share (TPS)

- **How to use:**
    - Students first **think** silently and write their answers.
    - Then, they **pair** up to discuss.
    - Finally, volunteers **share** responses with the class.
  - **Why it works:** Encourages individual thinking, peer interaction, and verbal expression.
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## 2. Visual Prompt Discussion

- **How to use:**
    - Project the image on the board.
    - Ask guiding questions like, “What kind of fire do you see here?” or “What would you do if you were Adam or Laila?”
    - Encourage open discussion before students write their answers.
  - **Why it works:** Builds observation skills and connects visuals to real-life thinking.
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## 3. Gallery Walk (Collaborative Reflection)

- **How to use:**
    - Print or post the image and questions in different areas.
    - Students walk around, read the scenarios, and add sticky notes with their answers.
    - Regroup and reflect on common responses.
  - **Why it works:** Promotes movement, peer learning, and critical thinking.
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## 4. Reflection Journals

- **How to use:**
    - After the discussion, students write their answers and thoughts in a personal reflection journal.
    - Prompts can include: “What surprised you?”, “What would you do in real life?”
  - **Why it works:** Helps internalize learning and track student growth over time.
- 

## 5. Interactive Notebook or Worksheet Completion

- **How to use:**
    - Students answer directly on printed worksheets like the one shown in the image.
    - Use as an **Exit Ticket** or for small-group check-ins.
  - **Why it works:** Structured, simple, and easy for assessment.
- 

## 6. Role Play or Scenario-Based Discussion

- **How to use:**
  - Divide students into groups and give each a scenario (from the image).
  - They role-play what they would do, then reflect on whether it was correct.
- **Why it works:** Enhances empathy, decision-making, and deeper understanding of safety procedures.

## 2. Explore

### Teacher's Role

- Facilitate hands-on or inquiry-based learning experiences.
  - Provide materials, images, or videos that show different fire types and scenarios.
  - Ask guiding questions like:
    - "What do you observe about these types of fires?"
    - "Do you think we can use water for all of them? Why or why not?"
  - Support students in making predictions or testing ideas (e.g., matching fire types with extinguishers).
  - Monitor group discussions and ensure all students are participating.
  - Avoid directly giving answers — instead, prompt students to investigate and discover.
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### Students Are Expected To

- Work in pairs or small groups to observe, analyze, or sort information about fire types or fire classes.
  - Ask questions and discuss their observations with peers.
  - Make connections between the image/story and real-world fire safety knowledge.
  - Record findings in a notebook or worksheet.
  - Begin identifying patterns or differences between types of fires (e.g., electric vs. grease fires).
  - Collaborate actively and respectfully.
- **Parts of the books included.**

Let's check this game to practice more about fire types and how to put them out.

<https://www.safetyapp.co.uk/appdemo>



- Give 2 minutes to think individually, 3 minutes to pair up and share.
- Then open a class discussion.

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#### 4. Resume Play and Complete Game

- Let students continue or finish their game, applying what they discussed in the reflection.
- Ask them to **focus on accuracy and decision-making** in the second half.

#### Strategy: Active Learning

- Students apply what they're learning immediately within the interactive environment.

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#### 5. Debrief & Group Wrap-Up

##### Teacher Role:

- Lead a **whole-class debrief** about what they learned:
  - "Which class of fire was hardest to identify?"
  - "Did the game change your thinking about fire safety?"
  - "How can this help us in real life?"

### **Strategy: Whole-Class Reflection + Metacognition**

- Encourage students to reflect on how their decisions changed based on new knowledge.
  - Use this discussion to reinforce key concepts.
- 

### **Materials Needed**

- Laptops, tablets, or computers with internet access.
- Headphones (optional for audio focus).
- Reflection sheets or notebooks.

### 3. Explain

#### Teacher's Role

##### 1. Guide Understanding

- Introduce both resources clearly, explaining what students should focus on (e.g., "As you watch, think about what fire needs to start" or "While reading, try to identify what causes each class of fire").

##### 2. Clarify Concepts

- Pause the video at key moments to highlight important vocabulary or clarify misconceptions (e.g., combustion, oxygen, ignition source).
- Summarize or discuss key points from the article with visual aids or anchor charts.

##### 3. Check for Understanding

- Ask follow-up questions such as:
  - "What are the three elements of fire?"
  - "What makes Class B fires dangerous?"
- Provide guiding questions or a note-taking graphic organizer to structure understanding.

##### 4. Link Back to Exploration

- Connect the video and reading to what they learned in the game or from the story of Adam and Laila.

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#### Student Expectations

##### 1. Engage with Multimedia Content

- Watch the video actively, possibly with a "watch and respond" sheet.
- Read the article to identify key terms and classifications.

##### 2. Take Notes or Complete a Graphic Organizer

- Fill in diagrams like the **Fire Triangle** or a **Fire Class Chart**.
- Write 1–2 sentences explaining each fire class in their own words.

##### 3. Participate in Discussion

- Answer and ask questions about what they viewed or read.
- Clarify confusions with peers or the teacher.

##### 4. Make Connections

- Relate new knowledge to earlier experiences in the story, game, or real life.

- **Parts of the books included.**



- **Watch It:**
  - *Before the video:*  
Introduce the purpose: “We’re going to learn what fire is and what keeps it burning.”  
Hand out a Fire Triangle graphic organizer.
  - *During the video:*  
Pause at key moments to:  
Ask questions like “Which part of the triangle is being discussed now?”  
Let students fill in the organizer with examples from the video.
  - *After the video:*  
Review the completed organizer as a class.  
Ask reflective questions: “Which element do you think is easiest to remove in real life?”
- **Read About It:**

**Print this table**

| Fire Class    | Fuel Type / Cause | Examples | How to Extinguish |
|---------------|-------------------|----------|-------------------|
| Class A       |                   |          |                   |
| Class B       |                   |          |                   |
| Class C       |                   |          |                   |
| Class D       |                   |          |                   |
| Class K(or F) |                   |          |                   |

### Instructions for Students:

1. While reading the article, **identify each fire class**.
2. Fill in:
  - What **kind of material** causes that class of fire (e.g., wood, oil, gas).
  - Real-life **examples** (e.g., kitchen fire, electrical fire).
  - The **correct way to extinguish** that type of fire (e.g., water, foam, CO<sub>2</sub> extinguisher).
3. After completing the table, discuss in groups or share your findings with the class.

#### 4. Elaborate

### 1. Focus (Multiple Questions to Answer)

- **Teacher's Role:**
  - Provide clear instructions and expectations for the questions.
  - Ensure the questions challenge students to reflect on what they've learned and apply it to new contexts.
  - Offer support and clarification if needed but encourage independent thinking.
  - Assess the students' understanding through their responses, providing immediate feedback when appropriate.
- **Expectations for Students:**
  - Answer a series of questions that test their understanding of the concepts.
  - Apply critical thinking and reasoning to explain their answers.
  - Show evidence of their learning by citing examples or previous lessons.

#### Parts of the book



1. What type of fire is Class A?

A) Electrical fires.  
B) Flammable liquids.  
C) Ordinary combustibles like wood and paper.  
D) Metal fires.

2. Which class of fire involves flammable liquids such as gasoline and oil?

A) Class A.  
B) Class B.  
C) Class C.  
D) Class D.

3. Class C fires involve:

A) Flammable metals.  
B) Electrical equipment.  
C) Cooking oils and fats.  
D) Ordinary combustibles.

4. What is the primary hazard of a Class D fire?

A) High heat and intensity.  
B) Electrical shock.  
C) Toxic fumes.  
D) Explosive reactions.

## **2. Practice (Poster Making)**

- **Teacher's Role:**
  - Explain the task of creating the poster, emphasizing the creativity and knowledge integration involved.
  - Encourage students to use various materials or digital tools to express their ideas effectively.
  - Monitor the students as they work, offering feedback or suggestions when necessary.
  - Foster collaboration if students are working in pairs or groups, facilitating discussions and brainstorming.
- **Expectations for Students:**
  - Use their knowledge to create a visual representation of what they've learned.
  - Demonstrate understanding of the topic through clear, engaging content on the poster.
  - Ensure the poster is visually appealing, informative, and includes key concepts or themes.



## Let's create a Fire Safety Poster using canva.



“Leave it to students creativity”

### 3. Challenge (Bonus Task)

- Teacher's Role:**
  - Provide the challenging question as an optional bonus task for students to work on at home.
  - Clarify any doubts or questions regarding the challenge but let students work independently on it.
  - Offer an opportunity for students to present or discuss their findings, providing encouragement and motivation.
- Expectations for Students:**
  - Independently attempt the challenging question at home.
  - Demonstrate higher-order thinking and problem-solving skills to answer the question.

- Use their creativity and understanding of the topic to explore complex ideas, even if they need to do additional research.
- Optionally, submit their answers to earn extra credit or recognition.

**Answer:**

Different fires burn with flames of different colors due to the **temperature of the fire** and the **chemical composition** of the materials being burned. Here's a breakdown of why this happens:

**1. Temperature:**

- The color of the flame is closely related to its temperature. For example, cooler flames (like those from a candle) appear yellow or orange, while hotter flames (like those from a welder's torch) can appear blue or even white.
- As the temperature increases, the color shifts from red to orange, yellow, and then to blue and white. Blue flames indicate a higher temperature and a more complete combustion.

**2. Chemical Composition:**

- Different materials release different elements when burned, and these elements emit specific colors when they are heated. For example:
  - **Sodium** burns with a bright yellow flame.
  - **Copper** can produce a green or blue flame.
  - **Strontium** produces a red flame.
  - **Lithium** burns with a red-pink flame.
- These different colors are a result of the energy released when electrons in the atoms of these elements move between energy levels, emitting light of specific wavelengths (colors).

## ❖ Relation to Standards:

### 1. Next Generation Science Standards (NGSS)

- **Science and Engineering Practices:**
  - **Planning and carrying out investigations:** Students might conduct an experiment (safely) to observe the color of flames from different materials, linking to investigative skills.
  - **Analyzing and interpreting data:** Students can analyze the data they gather from the flame color experiments and interpret it in the context of material composition and temperature.
- **Disciplinary Core Ideas:**
  - **PS1: Matter and Its Interactions:**
    - "The atomic composition of substances and their interactions can explain many properties of matter."
    - This lesson connects to understanding how the atomic structure of different materials affects their flame color during combustion.
  - **PS3: Energy:**
    - "Energy is transferred in many ways."
    - Understanding how heat energy affects the color of flames connects to energy transfer and temperature, which are fundamental concepts in physics.

### 2. Common Core State Standards (CCSS)

- **CCSS.ELA-LITERACY.RST.6-8.3:**
  - "Follow precisely a multistep procedure when carrying out experiments or tasks."
    - In the experiment, students would follow a procedure to observe and document the flame colors from various materials, aligning with this standard for detailed instructions and analysis.
- **CCSS.MATH.CONTENT.6.SP.B.5.C:**
  - "Summarize numerical data sets in relation to their context."
    - If students were to quantify the temperature of different flames or create charts, they would be summarizing data related to their flame color experiments.

### 3. National Science Education Standards (NSES)

- **Physical Science:**
  - "Position and motion of objects" and "properties of matter."
    - Understanding the chemical properties of materials and how they burn relates to the NSES standards on matter and energy transformations.

### 4. Safety Standards

- **Occupational Safety and Health Administration (OSHA) Standards:**
  - Discussing fire safety and different classes of fires (A, B, C, D) ties into real-world fire safety protocols and OSHA standards regarding the handling of hazardous materials and fire risks in workplaces.

## **5. International Baccalaureate (IB) Standards (if applicable)**

- **Science: Physical and Chemical Changes:**
  - Students can investigate the physical and chemical changes involved in combustion and the energy released during burning. The IB framework encourages inquiry-based learning and exploration, making it relevant to this lesson.

## **6. Environmental Standards**

- **Sustainable Development Goals (SDGs):**
  - **SDG 9 (Industry, Innovation, and Infrastructure) and SDG 12 (Responsible Consumption and Production):**
    - Understanding how fire interacts with different materials could lead to discussions on sustainable practices in fire safety and energy use, emphasizing the need for safety standards in industrial settings.

Duration: 45 mins

### 1- Engage

## Story

### 1. Teacher's Role in the Engagement Phase

- **Introduce the Scenario:**

The teacher will introduce the story about Adam and his friend deciding to create a smart fire truck for the STEAM fair, using it to hook students into the topic. The teacher should emphasize the excitement of the challenge ahead and make connections to real-world applications, such as fire trucks using technology to enhance fire-fighting efficiency.

- **Set Context and Purpose:**

- Explain the **purpose** of the lesson—students will be learning how to apply the **Engineering Design Process (EDP)** to design and prototype a **smart fire truck**.
- Show how this aligns with **STEAM principles** (Science, Technology, Engineering, Art, and Math).
- Encourage students to think about the problem-solving aspects, creativity, and teamwork involved in this project.

- **Pose Inquiry Questions:**

- Ask students open-ended questions that spark curiosity, such as:
  - “What kind of features might make a fire truck ‘smart’?”
  - “How can technology help firefighters do their job more efficiently?”
  - “What is the Engineering Design Process, and how can we use it to create our smart fire truck?”

- **Encourage Participation:**

- Make sure every student feels included by calling on different students to share their thoughts or ideas.
- Promote enthusiasm and engagement, ensuring that students are excited and ready to get involved in the project.



When Adam and Laila returned to school, they began sharing stories with their friends about what happened at the camp.

Later that day, Mrs. Sara announced the science fair to the students. Suddenly, an idea popped into Adam's mind—he decided to make a smart fire truck for his project and asked Laila to work with him.



## Model EDP:

- Briefly introduce the **Engineering Design Process (EDP)** by explaining the stages:
  1. **Ask** (Identify the problem).
  2. **Imagine** (Brainstorm ideas).
  3. **Plan** (Make a design plan).
  4. **Create** (Build a prototype).
  5. **Test and Improve** (Evaluate and refine the design).

[https://www.youtube.com/watch?v=MAhpfFt\\_mWM](https://www.youtube.com/watch?v=MAhpfFt_mWM)

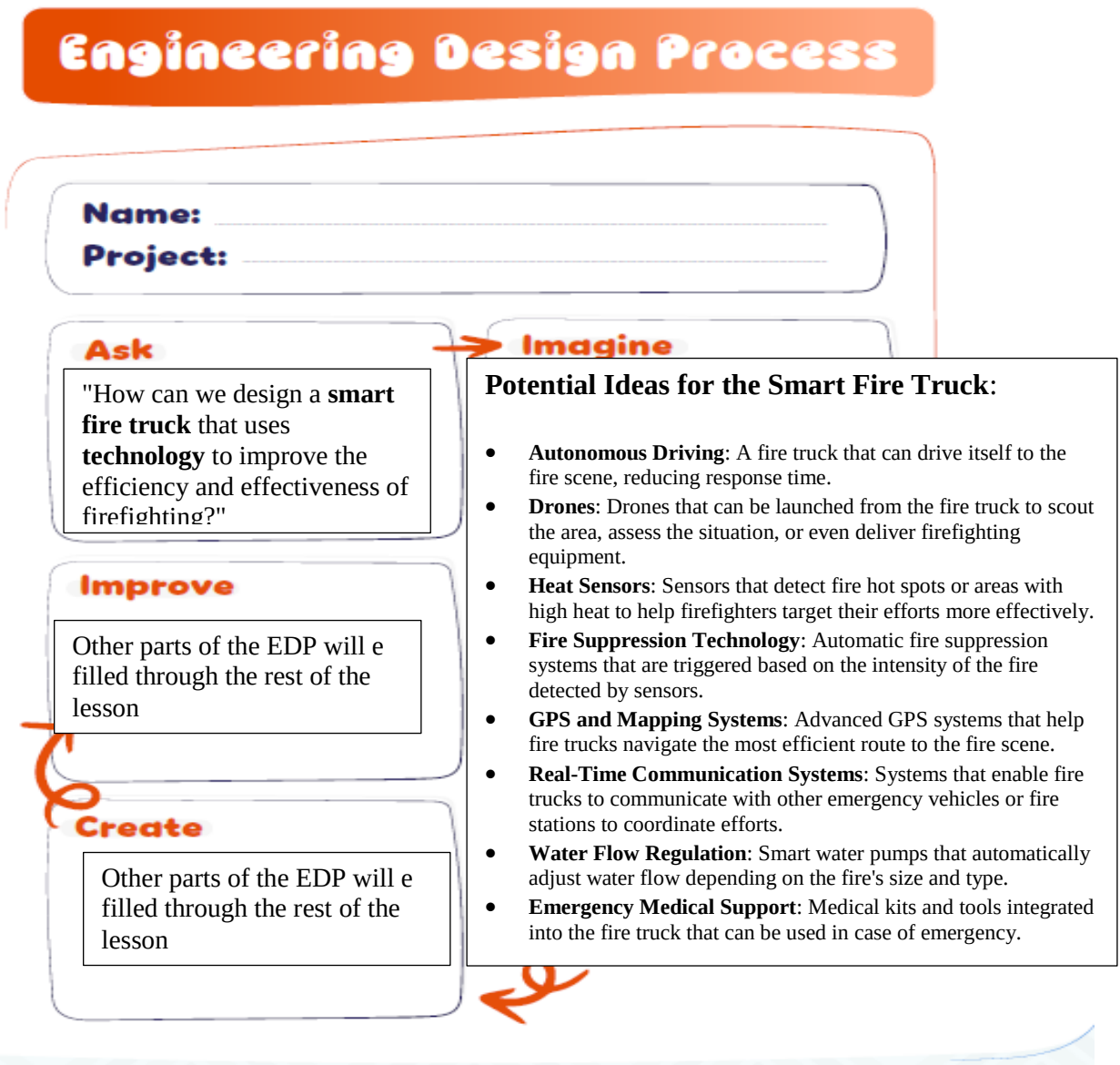
## 2. Expectations from Students

- **Engagement and Participation:**
  - Students are expected to actively listen to the story, engage with the teacher's questions, and contribute ideas about how they would approach designing a smart fire truck.
  - They should start thinking about the role of **technology** in a fire truck and how it could be made smarter, considering various aspects like automation, sensors, communication, etc.
- **Critical Thinking:**
  - Students should demonstrate curiosity about the problem (smart fire truck) and ask questions to understand it better.
  - They are expected to begin to think about the **requirements** of a fire truck and how they might incorporate technology to enhance it.
- **Collaboration:**
  - Students will begin working with peers, either in pairs or small groups, to brainstorm ideas for the smart fire truck, applying the first steps of the **Engineering Design Process**.
- **Reflective Thought:**
  - Encourage students to reflect on the lesson's goal and how the EDP will guide their project.
  - They may also start sketching ideas or writing down their initial thoughts.

### Strategies to Implement

- **Interactive Storytelling:**
  - Use storytelling to engage students, like how the teacher presents Adam's journey and his excitement about the STEAM fair. This can inspire students and set a fun tone for the lesson.
  - Include visual aids, like images of fire trucks, STEAM fairs, or the Engineering Design Process, to enhance understanding.
- **Think-Pair-Share:**
  - Ask students to think about their initial ideas for a smart fire truck, then discuss them with a partner, and finally share them with the class. This strategy ensures that all students are actively involved and that quieter students have a chance to contribute.
- **Brainstorming Session:**
  - Organize a **brainstorming session** where students generate as many ideas as possible for features that could make a fire truck "smart." Encourage wild and creative ideas (e.g., drones, sensors, automatic fire suppression systems).

- Write all ideas on the board and let students vote on which ones they find most interesting or feasible.
- **Use of Visual Aids:**
  - Provide visual prompts like diagrams of fire trucks or videos showing modern smart technologies used in fire trucks. This helps students visualize the challenge they are tackling.
- **Link to Real-World Examples:**
  - Show real-life examples of smart technologies used in fire trucks, such as autonomous vehicles, GPS systems, sensors, or communication systems that help firefighters. This can spark students' interest and give them concrete examples to work with.
- **Provide Frameworks:**
  - Introduce a **simple rubric or outline** for the Engineering Design Process that students can follow throughout the lesson and project. This will help students focus on the steps they need to take and guide them through the process.



## 1. Ask (Identify the Problem)

In this phase, students will need to **define the problem** and gather any necessary background information. The goal is to understand the challenge and its requirements.

- **Problem Statement:**  
"How can we design a **smart fire truck** that uses **technology** to improve the efficiency and effectiveness of firefighting?"
- **Guiding Questions for Students:**
  - **What are the essential functions of a fire truck?**
  - **What challenges do fire trucks face during firefighting operations?**
  - **How can technology help solve these challenges?**
  - **What are the specific technologies that might make a fire truck 'smart' (e.g., sensors, automation, communication systems)?**
  - **How can we design a system that helps firefighters respond faster, be safer, or more effective in putting out fires?**
- **Learning Outcomes (for the Ask phase):**
  - Students should be able to identify the problem (smart fire truck design).
  - Understand the real-world needs of firefighters and the role of technology in firefighting.
  - Define the features that a smart fire truck should have.

## 2. Brainstorming (Imagine)

In this phase, students will generate a wide variety of ideas and **solutions**. The goal is to think creatively and consider all possibilities.

- **Brainstorming Guidelines:**
  - Encourage students to **think creatively** and come up with as many ideas as possible, even if some seem impractical at first.
  - Foster an environment where **all ideas are welcome** and that no idea is too outlandish.
  - Ask students to consider a combination of technologies, such as **automation, sensors, and communication**, that might work together.
- **Potential Ideas for the Smart Fire Truck:**
  - **Autonomous Driving:** A fire truck that can drive itself to the fire scene, reducing response time.
  - **Drones:** Drones that can be launched from the fire truck to scout the area, assess the situation, or even deliver firefighting equipment.
  - **Heat Sensors:** Sensors that detect fire hot spots or areas with high heat to help firefighters target their efforts more effectively.
  - **Fire Suppression Technology:** Automatic fire suppression systems that are triggered based on the intensity of the fire detected by sensors.
  - **GPS and Mapping Systems:** Advanced GPS systems that help fire trucks navigate the most efficient route to the fire scene.
  - **Real-Time Communication Systems:** Systems that enable fire trucks to communicate with other emergency vehicles or fire stations to coordinate efforts.
  - **Water Flow Regulation:** Smart water pumps that automatically adjust water flow depending on the fire's size and type.



- **Emergency Medical Support:** Medical kits and tools integrated into the fire truck that can be used in case of emergency.
- **Tools for Brainstorming:**
  - **Mind Mapping:** Have students create a mind map where they write the main idea (smart fire truck) in the center and branch out with different technologies and solutions that might be part of the design.
  - **Group Discussion:** Organize students into small groups where they can discuss and build on each other's ideas. Afterward, each group shares their ideas with the whole class.
  - **Idea Voting:** After brainstorming, students can vote on the most feasible and exciting ideas to pursue further.

### **Outcome of the Ask and Brainstorming Phase:**

- **Ask:** The problem is clearly defined: creating a **smart fire truck** using **technology** to enhance firefighting.
- **Brainstorming:** Students generate a variety of creative ideas, considering different technologies and how they can be applied to improve the design and function of a fire truck.

**Note:** The rest of the EDP will be filled step by step through the rest of the theme.

# Mechanics Domain

LESSON 03  
Get the water out.

Duration: 90 mins

## Stepping Stone:

### You need to know ...



Simple mechanical concepts such as levers, gears, and pulleys.



Definition of hydraulic force.

### You will be able to...



Understand how pressure affect water coming out from the truck.



Determine the best diameter for the hose.

## 1-Engage

### Story

## 1. Teacher's Role

- **Facilitator:**

- The teacher should **guide** the students through the process of understanding the problem and thinking critically about what they need to include in their design. The teacher helps students focus on the essential components of the fire truck and how they might approach the design challenge.
- **Ask open-ended questions** to help students reflect on what is necessary to include in their design (e.g., "What basic parts do you think are important for a fire truck to have?" "How can we make sure the water comes out in an efficient way?").
- **Provide context and real-world examples** of fire trucks and firefighting equipment, showing students how complex designs work in real life.
- **Encourage group discussion:** The teacher should encourage students to discuss their ideas with peers, fostering collaboration and teamwork.
- **Connect to Engineering Design Process (EDP):** Ensure that students understand the next steps of the EDP, and encourage them to begin considering the "Ask" and "Imagine" phases as they define the essential elements of the smart fire truck.

## 2. Expected from Students

- **Initial Problem-Solving:**

- Students should begin identifying key **parts** of the smart fire truck they think are essential for the design, such as the water system and other basic components like wheels, pumps, or sensors.
- They will **discuss and share** ideas about how to solve the problem of how water will come out and which parts should be included in the fire truck's design.

- **Collaboration:**
  - Students are expected to **work together** to develop their ideas and help each other narrow down their design choices.
  - They will **ask questions**, consider different perspectives, and start thinking critically about the features they want to include.
- **Conceptual Thinking:**
  - Students should think about the **basic mechanical and technological components** that make up a fire truck and how they can make these parts smarter, more efficient, or more effective.
  - Students will also need to think about the **functionality** of each component (e.g., water delivery system) and how these parts will work together in the design.

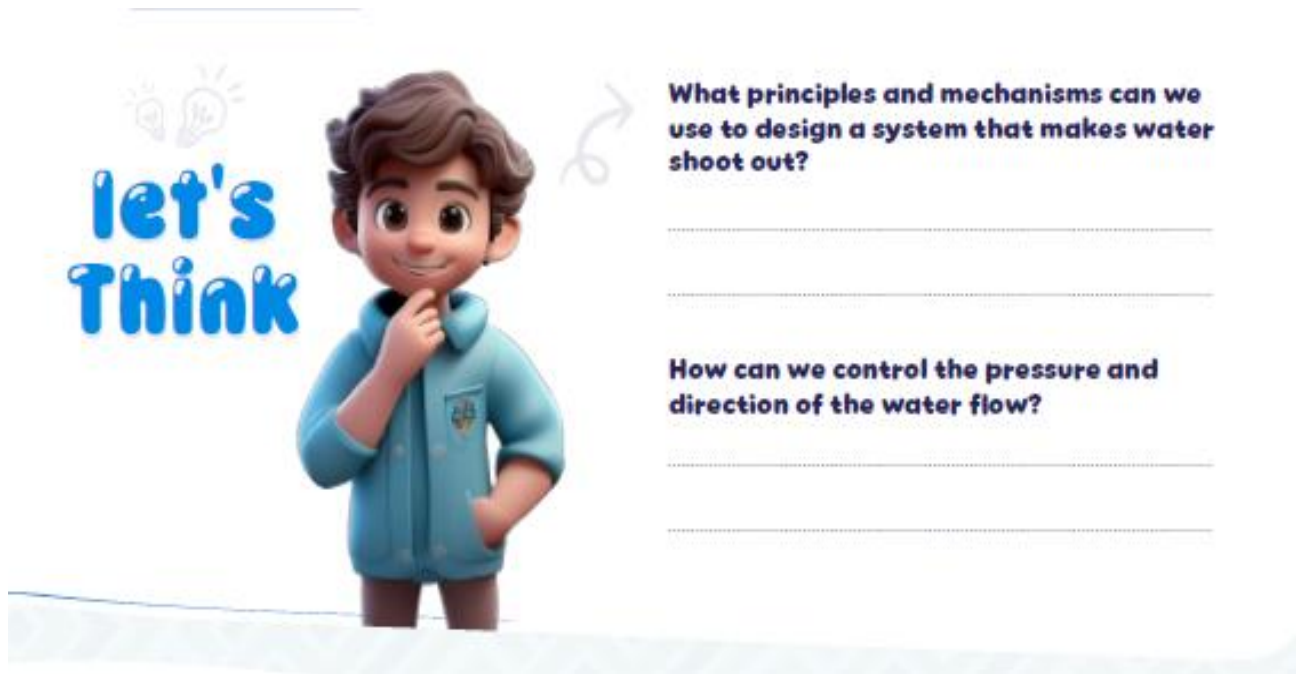
## Implementation Strategies

To implement the engagement phase effectively, the teacher can use the following strategies:

- **Think-Pair-Share:**
  - Ask students to think individually about the problem for a few minutes, then pair up with a partner to discuss their ideas about what basic parts are needed for the fire truck. Afterward, ask each pair to share their thoughts with the class.
  - This encourages **individual thinking** followed by **collaborative discussion**.
- **Guided Group Discussion:**
  - After introducing the problem, facilitate a class-wide discussion where students can share their ideas about the basic parts of a fire truck. Use **prompting questions** to guide them toward thinking about the critical systems involved (e.g., water delivery system, wheels, pumps, sensors, etc.).
- **Visual Aids:**
  - Use **diagrams or pictures** of fire trucks to visually represent the essential parts of a fire truck. This helps students connect their ideas with real-world examples and provides a starting point for their design brainstorming.
- **Brainstorming Session:**
  - Organize a **brainstorming session** where students can come up with different ideas for their fire truck. For instance, they can think about:
    - **Water delivery system** (manual or automatic, what mechanism will release the water, etc.)
    - **Sensors** that can detect the fire or measure water levels.
    - **Mechanical parts** like wheels, pumps, and hoses.
    - **Smart systems** such as automated driving or communication systems.
  - **Post-Its or Mind Maps:** Encourage students to use sticky notes or a large mind map on the board where they can add their ideas. This can visually show how different parts of the fire truck connect together.
- **Real-World Examples:**
  - Show real-life examples or videos of **fire trucks** and their features. This could include automated systems, fire-fighting equipment, and technologies used in modern fire trucks.
- **Reflection and Discussion:**

- After brainstorming, have students **reflect** on the ideas they generated and choose the most important components.
- Ask students: "Why do you think this part is important? How will it help your fire truck be more effective?"
- This helps students prioritize their ideas and focus on what is essential for the smart fire truck design.

## Think



### Possible answers for these questions

#### 1. What principles and mechanisms can we use to design a system that makes water shoot out?

- **Pressure Mechanism:**
  - **Water Pump:** A pump can push the water through pipes and force it out with pressure, making it shoot out of the nozzle. The stronger the pump, the farther the water can go.
  - **Air Pressure:** Using air to push water through a hose can also work, like how a balloon pops when air pressure builds up.
- **Mechanical Mechanisms:**
  - **Trigger or Lever:** A lever or trigger can be used to start the water flow, like how a garden hose nozzle works when you squeeze it.
  - **Valve:** A valve controls the water flow, turning it on and off. It can be like the tap in your house—open it to let water flow, and close it to stop.
- **Nozzle Design:**

- **Adjustable Nozzle:** A nozzle can be designed to control how wide or narrow the water shoots out. Some nozzles have different settings to spray water in a wide mist or a strong, focused stream.

## 2. How can we control the pressure and direction of the water?

- **Pressure Control:**
  - **Pressure Regulator:** A pressure regulator helps control how strong the water comes out. If the water pressure is too high, it could be hard to handle, and if it's too low, it won't put out the fire well.
  - **Pump Speed:** Changing how fast the pump works can change the pressure. A faster pump makes water come out with more force, while a slower pump gives less pressure.
- **Direction Control:**
  - **Rotating Nozzle:** The nozzle can spin to change the direction of the water, allowing you to aim it where it's needed most.
  - **Tilt Mechanism:** The nozzle can be moved up or down to control the angle of the water stream, helping you target different areas.
- **Manual Control:**
  - **Levers and Knobs:** Firefighters can use levers or knobs to control the water flow and nozzle direction manually. For example, turning a knob could make the water spray in a wide mist, or using a lever could make the water shoot in a straight line.

## 2- Explore

### 1. Teacher's Role in the Explore Phase:

- **Facilitator:**
  - The teacher should set up the simulation, explain how to use it, and provide students with clear instructions on how to interact with the various controls (e.g., adjusting pressure, flow, or pipe size).
- **Guide Students' Exploration:**
  - As students interact with the simulation, the teacher should walk around and **observe** how students are experimenting. Offer help if needed but encourage students to explore on their own and think critically about the results they are seeing.
- **Encourage Critical Thinking:**
  - The teacher should **ask guiding questions** to help students reflect on their findings. For example:
    - "What happens when you increase the pressure?"
    - "How does the size of the pipe affect the flow of water?"
    - "Can you identify what changes the direction of the water in this simulation?"
- **Connect Simulation to Real-World Application:**
  - The teacher should emphasize how this simulation relates to the **smart fire truck** they are designing. Encourage students to think about how pressure is important in the real-world systems they are creating.



Let's use this simulation to learn more about how pressure works.

<https://phet.colorado.edu/sims/cheerful-fluid-pressure-and-flow/cheerful-fluid-pressure-and-flow.html?simulation=fluid-pressure-and-flow>



## Expected from Students:

- **Active Exploration:**
  - Students are expected to **actively engage** with the simulation by experimenting with different settings for pressure, pipe size, and flow. They should explore the effects of changing these variables.
- **Problem Solving:**
  - Students should try to **solve problems** they encounter while interacting with the simulation. For example, they may experiment with different combinations of pressure and pipe size to observe how they affect the flow of water.
- **Observation and Data Recording:**
  - Students should **observe** the results of their experiments carefully and **record their findings**. They should note how pressure and flow are related and the impact of pipe size on the flow.
- **Make Connections to Fire Truck Design:**
  - Students should relate what they learn in the simulation to the design of the smart fire truck, thinking about how pressure and flow are important for controlling the water system on their truck.

## Steps for Implementation

1. **Introduce the Simulation:**
  - Start by introducing the simulation to the students. Explain the basic controls and how the simulation demonstrates the flow of water under different conditions (pressure, pipe size, etc.).
  - Ensure students understand the purpose of the activity: to explore how pressure and flow affect the system.
2. **Initial Exploration:**
  - Give students time to freely experiment with the simulation. Let them adjust the pressure, pipe sizes, and observe the outcomes. Encourage them to explore all the settings.
3. **Guided Observation and Reflection:**
  - Ask students to observe what happens when they adjust different variables. Guide them through the process of recording their observations (e.g., "What happens when pressure is increased?" "How does the water flow change with different pipe sizes?").
4. **Small Group Discussions:**
  - After some initial exploration, have students discuss their observations in small groups. Encourage them to share what they've discovered and come up with answers to key questions about how pressure works.
5. **Make Connections:**
  - Have students connect what they learned from the simulation to their fire truck design. Ask them to consider how **pressure** and **water flow** could affect their design and how they might control these variables in their project.
6. **Wrap-up and Reflection:**
  - Bring the class together to discuss their findings. Encourage students to reflect on how understanding pressure and flow might help them design a better water delivery system for their fire truck.

## Strategies to Implement

- **Guiding Questions:**

Ask specific questions during the exploration to guide thinking:

- “What happens when you increase the pressure? Why do you think that happens?”
- “How does the size of the pipe change the flow of water?”
- “How can we use this information to design the water delivery system on our fire truck?”

- **Pair and Share:**

After students explore, have them pair up with a classmate to discuss their findings and share insights. This helps reinforce concepts through peer collaboration.

- **Think-Aloud:**

Encourage students to **think aloud** as they interact with the simulation. Have them explain what they are doing and why they think certain changes affect the outcome. This promotes deeper thinking.

- **Interactive Learning:**

Use an interactive class discussion at the end of the activity where students present their findings from the simulation. Discuss how pressure and flow can be controlled in real-world systems, such as fire trucks.

- **Real-World Examples:**

Relate the findings in the simulation to real-world systems, such as fire truck water pumps. This helps students see the practical application of the concepts they are learning.



## 3- Explain

### Teacher's Role in the Explain Phase:

- **Provide Clear Instructional Content:**

Select and show a video that visually and clearly explains the **function of fire trucks**, especially how the **water spraying mechanism** works (e.g., pump, hose, water tank, nozzles).

- **Bridge Prior Learning to New Knowledge:**

Link what students explored in the simulation (fluid pressure and flow) to what they're seeing in the video.

- **Pause and Discuss:**

Pause the video at key points to ask questions or emphasize parts (e.g., "Why is the pump important?" or "What allows the water to reach tall buildings?").

- **Clarify Misconceptions:**

If students misunderstood any parts during the Explore phase, now is the time to clarify those concepts with accurate vocabulary and visuals.

- **Encourage Reflection:**

Ask students to reflect on what they learned and **summarize** key ideas in their own words.

## What Is Expected from Students?

- **Active Listening and Observation:**  
Watch the video attentively to understand how the system works in real life.
- **Note-Taking:**  
Write down key parts of the fire truck's system: the pump, the tank, the hose, the nozzle, and how water is pressurized and directed.
- **Compare and Connect:**  
Connect the explanations in the video to their previous exploration (e.g., how pressure is created and directed through pipes).
- **Ask and Answer Questions:**  
Engage in teacher-led discussions by asking questions and responding to prompts.
- **Use Scientific Terms:**  
Begin using correct vocabulary (e.g., **valve, pressure, hose, nozzle, tank, pump**) when describing how the fire truck works.

Video link: <https://www.youtube.com/watch?v=C7wy3vWbMjg>

### Strategies to Implement

#### 1. Pre-Video Discussion

Ask students:

- "What do you already know about how fire trucks spray water?"
- "What do you want to learn from this video?"

#### 2. Video Pause-and-Think

Pause at moments where key mechanisms are shown.

Ask guiding questions like:

- "What's happening here?"
- "How do you think the water gets pressurized?"
- "Which part of the fire truck is responsible for water direction?"

#### 3. Video Worksheet or Graphic Organizer

Provide a fill-in-the-blank sheet or a labeled diagram of a fire truck system to complete while watching.

Examples:

- "Water comes from the \_\_\_\_."
- "The \_\_\_\_ increases water pressure."

#### 4. Think-Pair-Share

After the video, ask a question like:

- "How does the fire truck make the water shoot out with force?"  
Let students think, discuss with a partner, then share ideas with the class.



## 5. Class Discussion with Real-World Connection

After the video, hold a class discussion where students describe the water-spraying process using what they've learned from both the simulation and the video.

### 4- Elaborate

#### Focus

#### Teacher's Role:

- Prepare and present the quiz (digitally or on paper).
- Guide students through instructions and ensure they understand how to answer.
- Clarify any doubts before students begin.
- Collect, observe, or review answers to identify understanding gaps.
- Use student performance to decide if further explanation is needed.

#### Students' Role:

- Read and analyze each question carefully.
- Apply their understanding of fluid pressure and fire truck mechanics to select the correct answers.
- Use scientific reasoning and vocabulary to support their choices if asked to justify.
- Reflect on their understanding through the questions.

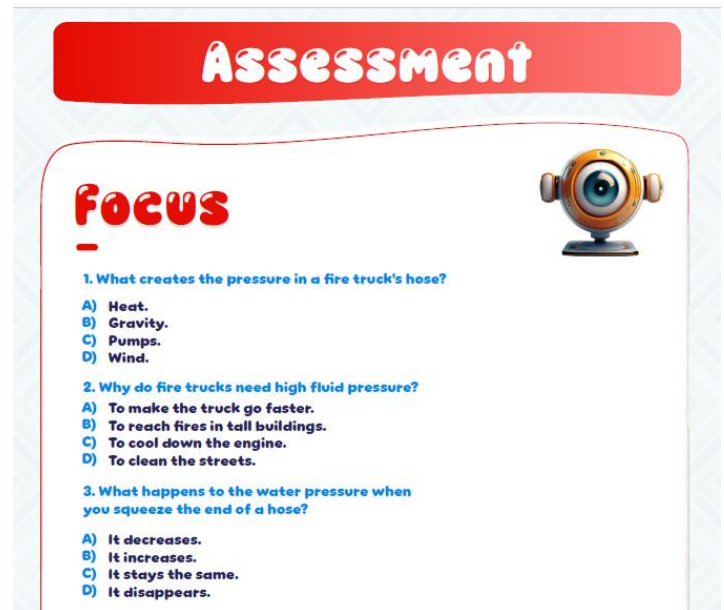
#### Questions Answer

- What creates the pressure in a fire truck's hose?

A) Heat  
B) Gravity  
C) Pumps  
D) Wind

- Why do fire trucks need high fluid pressure?

A) To make the truck go faster  
B) To reach fires in tall buildings  
C) To cool down the engine  
D) To clean the streets



- What happens to the water pressure when you squeeze the end of a hose?

- A) It decreases
- B) It increases
- C) It stays the same
- D) It disappears

## Practice



### Activity Prompt:

“Select the best diameter for your hose, choose the right material, and begin creating it.”

### Teacher’s Role:

- Provide students with a set of materials (plastic tubing in different diameters, connectors, cardboard, tape, rubber sheets, etc.).
- Facilitate discussion around factors affecting water flow (diameter, material, length).
- Guide students as they brainstorm and test options.
- Ask probing questions like:
  - “Why did you choose this diameter?”
  - “How does the material affect water flow or pressure?”
- Ensure safety during the building/testing phase.

### Students’ Role:

- Apply their knowledge from the previous phases to design and build a hose.
- Select the diameter and material based on reasoning and observations.
- Construct the hose model as part of the smart fire truck project.
- Test and refine their design if time allows.
- Work collaboratively in teams and document their decision-making process.

### Steps for Practice Activity:

1. **Review Objective:**  
Revisit what they’ve learned about pressure, flow, and hose function.
2. **Material Selection:**  
Examine and compare different tubing options (diameter, flexibility, texture).
3. **Design Planning:**  
Decide on the best diameter and material. Sketch or outline the hose plan.
4. **Construction:**  
Cut, connect, and build the hose using provided materials.
5. **Integration:**  
Attach the hose to their fire truck prototype (if available), or simulate how it would connect.

6. **Optional Testing (if time and materials allow):**

Simulate or observe water flow through the hose.

7. **Reflection:**

Discuss what worked, what didn't, and why their design is effective.

Print this worksheet to help you with the activity:

<https://docs.google.com/document/d/118GVgaBHW5A4Y5JAfkgVbV55CuDyjeT7AUay-sElHE0/edit?usp=sharing>

**Let's return to our  
EDP paper and  
update the planning  
section.**



## Planning

After researching how fluid pressure works and exploring the effects of hose diameter and materials, we decided to:

- Use a **medium-diameter rubber hose** to balance water flow and pressure.
- Include a **small nozzle opening** to increase water pressure and spraying distance.
- Add a **simple manual or motorized pump** system to control the water pressure.
- Design the nozzle with a **movable joint** to allow directional control of the spray.

## Relation to Standards

### NGSS – Next Generation Science Standards

#### 1. MS-ETS1-1:

*Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution.*

- In your lesson, students define how to design a hose that meets criteria like strong water pressure, correct material, and proper direction control.

**2. MS-ETS1-2:**

*Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.*

- Students test materials, compare hose diameters, and justify their final design choices.

**3. MS-PS3-3:**

*Apply scientific principles to design, construct, and test a device that either minimizes or maximizes thermal energy transfer.*

- While your lesson focuses on water pressure, students indirectly learn about energy transfer through pumps and hose behavior.

**4. MS-PS2-2:**

*Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object.*

- Understanding water flow and pressure supports this standard in context.
- 

**ISTE Standards for Students (Technology)****1. Innovative Designer (4a & 4d):**

- Students use a design process to generate ideas, test theories, create innovative artifacts, and solve authentic problems.
  - Students exhibit a tolerance for ambiguity, perseverance, and the capacity to work with open-ended problems.
- 

**Common Core State Standards (CCSS) Connections****ELA-LITERACY.RST.6-8.3:**

*Follow precisely a multistep procedure when carrying out experiments, taking measurements, or performing technical tasks.*

- Students conduct guided exploration and build a prototype using clear steps.

**ELA-LITERACY.WHST.6-8.7:**

*Conduct short research projects to answer a question (including a self-generated question), drawing on several sources and generating additional related, focused questions that allow for multiple avenues of exploration.*

- Students explore how pressure works and apply that to their fire truck system.

Duration : 90 mins

## Stepping Stone:

### You need to know ...



3D Design Concepts.

### You will be able to...



Understand and use fusion software.



Create 3D models of fire truck parts using Fusion 360 software.

1- Engage

## Story



Adam

Okay, I think we're making progress. Let's start designing the parts of the fire truck.

### Teacher Role

- Introduce the context/story to spark curiosity (e.g., recap Adam and Laila's conversation about designing the fire truck).
- Ask thought-provoking questions to guide student thinking.
- Observe student responses and encourage participation.
- Set a real-world challenge that students will solve using the EDP (Engineering Design Process).
- Avoid giving solutions—focus on generating interest and guiding inquiry.

### Expected from Students

- Actively listen to the story or scenario.
- Share prior knowledge about fire trucks, water pressure, or related concepts.
- Ask questions or express curiosity.
- Start to think about the challenge and possible ways to solve it.
- Collaborate with peers to discuss ideas.

## Strategies to Implement

### 1. Storytelling:

- Start with a short, engaging story involving characters (e.g., Adam and Laila).
- Connect the story to a real-world problem like fire safety or smart vehicles.

### 2. Question Prompts:

- Use open-ended questions such as:
  - “How do you think fire trucks spray water so far?”
  - “What might we need to build a smart fire truck ourselves?”
  - “What parts do you think are most important in the design?”

### 3. Think-Pair-Share:

- Give students time to think individually, then share ideas with a partner, and finally with the whole class.

### 4. KWL Chart (Know, Want to know, Learned):

- Fill in the “K” and “W” columns to activate prior knowledge and set learning goals.

### 5. Visual Stimuli:

- Show images or short clips of fire trucks in action to spark curiosity.

### 6. Group Brainstorming:

- Let students share what they think a smart fire truck needs or should do.

## Think

Here are some of the possible answers:

let's  
Think

What parts should we include  
in our fire truck?

- **Water Pump:** Essential for drawing water from the tank or external sources and pumping it through hoses to extinguish fires.
- **Fire Hose:** Flexible tubes that carry the water from the pump to the fire. Different nozzles can adjust the water flow and pressure.
- **Ladder:** Allows firefighters to reach higher floors or places where people may be trapped.
- **Aerial Platform or Ladder Truck:** A platform or extendable ladder mounted on the truck to reach significant heights, often used for rescues or fighting fires from above.
- **Siren and Lights:** Emergency sirens and flashing lights are used to alert others of the approaching truck and clear the way during emergencies.
- **Fire Extinguishers:** Smaller, portable extinguishers on the truck to quickly put out smaller fires that are manageable without the hose.
- **First Aid Kit:** Contains medical supplies to treat injuries and help those affected by the fire until paramedics arrive.

- **Rescue Tools:** Tools such as the jaws of life, axes, crowbars, and sledgehammers are used for breaking doors, windows, and rescuing people from accidents.
- **Water Tank:** A large tank that stores water on the truck when there are no nearby fire hydrants, allowing for immediate firefighting.
- **Thermal Imaging Camera:** A device that helps firefighters see through smoke to locate fire hotspots or trapped individuals.
- **Communication System:** Radios and communication tools to keep in contact with the fire station and other responders during an emergency.
- **Firefighter Protective Gear:** Includes helmets, jackets, gloves, boots, and masks to protect firefighters from heat, smoke, and other hazards.

## Strategies to implement

- **How it works:** Ask students to individually think about the parts of a fire truck for a couple of minutes. Then, have them pair up with a classmate to discuss their ideas. Afterward, each pair shares their answers with the whole class.
- **Purpose:** Encourages individual thinking followed by collaborative discussion, allowing students to reflect and refine their ideas before sharing.

### 2. Group Brainstorming

- **How it works:** Split the class into small groups and provide them with a large piece of paper or a digital document to list all the parts they think should be included in a fire truck. After 10-15 minutes, each group shares their list with the class.
- **Purpose:** Collaborative activity that encourages multiple perspectives and engages all students in the thinking process.

### 3. Guided Questioning

- **How it works:** Ask leading questions like:
  - "What do you think firefighters need to do when they arrive at a fire?"
  - "What equipment would be useful for rescuing people?"
  - "How would the truck help firefighters reach and put out the fire?" Encourage students to answer based on these prompts.
- **Purpose:** Helps students think critically about the practical functions of the fire truck parts and their necessity.

### 4. Interactive Polling or Voting

- **How it works:** After discussing the possible parts as a class, use a tool like Kahoot, Mentimeter, or a simple show of hands to vote on the most important parts of the fire truck.
- **Purpose:** Engages students in an interactive way, while also highlighting the importance of prioritizing features.

## 5. Role-Playing Activity

- **How it works:** Have students role-play as firefighters. Give them scenarios where they need to decide what equipment from the fire truck to use based on the situation. Ask them to justify their choices.
- **Purpose:** Allows students to think about the function of each part of the fire truck in real-life contexts, making the learning process more relatable.

## 6. Use of Visual Aids

- **How it works:** Show pictures, diagrams, or videos of fire trucks and ask students to identify the parts. Afterward, they can discuss the purpose of each part.
- **Purpose:** Visual aids help students better understand and recall the different components of the fire truck.

## 7. Class Discussion

- **How it works:** Lead a class discussion, asking students what they know about fire trucks and why certain parts are essential. As they contribute, write their ideas on the board.
- **Purpose:** Facilitates a structured conversation where students can learn from one another and refine their thoughts.

2- Explore

### Teacher Role

- Introduce the AI image generator tool and explain how it will be used to design a fire truck.
- Set expectations for students: they will create an image of a fire truck with key parts based on their learning.
- Ask guiding questions like: “What are the essential parts of a fire truck?” and “How does each part contribute to its function?”
- Observe student engagement and offer assistance when needed as they experiment with the AI tool.
- Encourage reflection by asking students to explain their design choices and the importance of each part.
- Avoid providing solutions—focus on facilitating exploration and creativity.

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### Expected from Students

- Actively engage with the AI image generator, inputting descriptions of fire truck parts based on prior knowledge.
- Experiment with different prompts to explore various fire truck designs.
- Justify the design choices made by explaining how each part serves a functional purpose in the fire truck.
- Collaborate with peers to discuss and refine their ideas.
- Reflect on the created image and share insights on why certain parts were included in the fire truck design.



In this part instructors will allow the students to write the parts they discussed to image generating AI and help them to write the correct prompt.  
They can use : ChatGpt – Freepik – Canva – Gemini.

3- Explain



**Use the AI image generator to visualize how your fire truck will look.**

**Follow the link:**

<https://copilot.microsoft.com/>



## Teacher Role

- Provide students with an article or video that explains key concepts related to fire trucks and their parts (e.g., how they work, the function of different components).
- Explain the content of the article or video, breaking it down into manageable sections.
- Guide a class discussion, prompting students with questions to ensure they understand the material and can relate it to their own designs.
- Encourage students to connect the information from the article/video with their previous work (e.g., the fire truck designs they created using the AI tool).
- Facilitate the discussion by asking open-ended questions like: “What part of the fire truck surprised you the most?” and “How do the parts you added in your design contribute to its effectiveness?”
- Summarize key points and ensure students grasp the core concepts.

---

## Expected from Students

- Actively watch the video or read the article, taking notes on key information about fire trucks and their parts.
- Participate in the class discussion, sharing insights from the article or video.
- Relate the content of the article or video to their own ideas and designs, explaining how the new knowledge helps refine their understanding.
- Ask questions to clarify any confusion and deepen their understanding.
- Engage with peers in the discussion, listening to different perspectives and contributing to the group’s learning process.

## Watch it

You can use the previous strategies mentioned in this part.

## Read it



### 1. Pre-Reading Activity

- **How it works:** Ask questions to activate prior knowledge (e.g., “What do you know about fire trucks?”).
- **Purpose:** Set the stage for the reading by tapping into existing knowledge.

### 2. Guided Reading

- **How it works:** Pause during the reading to discuss key points and check for understanding.
- **Purpose:** Ensure comprehension and guide students through the article.



### 3. Jigsaw Method

- **How it works:** Split the article into sections for small groups to read and then share with the class.
- **Purpose:** Promote collaboration and break down the reading into manageable parts.

### 4. Think-Pair-Share

- **How it works:** Have students think about a question, discuss it with a partner, and then share with the class.
- **Purpose:** Foster discussion and ensure all students engage with the content.

### 5. Annotating the Article

- **How it works:** Students underline key points and write questions or comments in the margins.
- **Purpose:** Encourage active reading and reflection on the content.

### 6. Collaborative Note-Taking

- **How it works:** Have students take notes together on a shared document or chart paper.
- **Purpose:** Promote teamwork and create a collective reference point.

### 7. Visual Aids and Diagrams

- **How it works:** Provide diagrams related to the article to aid understanding.
- **Purpose:** Support comprehension with visual representation.

## 8. Question-Answer Relationships (QAR)

- **How it works:** Ask questions based on the article, from literal to higher-order thinking.
- **Purpose:** Develop reading comprehension and critical thinking.

## 9. Summarization

- **How it works:** Students summarize the article in their own words.
- **Purpose:** Help students consolidate understanding.

## 10. Think-Alouds

- **How it works:** Teacher models thinking aloud while reading parts of the article.
- **Purpose:** Demonstrate how to analyze and interpret the text.

## 11. Peer Teaching

- **How it works:** Students explain key points to a peer.
- **Purpose:** Reinforce understanding through teaching.

Elaborate and Evaluate

### Teacher Role

- Guide students through reflecting on their fire truck designs and the concepts learned, prompting them to assess their work based on functionality, design, and problem-solving.
- Ask probing questions (e.g., "Why did you choose this part for your fire truck?") to help students justify and explain their design choices.
- Provide constructive feedback and encourage students to make revisions based on their reflections and feedback.
- Challenge students to extend their thinking by applying the concepts to new or more complex scenarios (e.g., designing a fire truck for a mountain town).
- Facilitate group discussions where students can share and elaborate on ideas, expanding their designs.

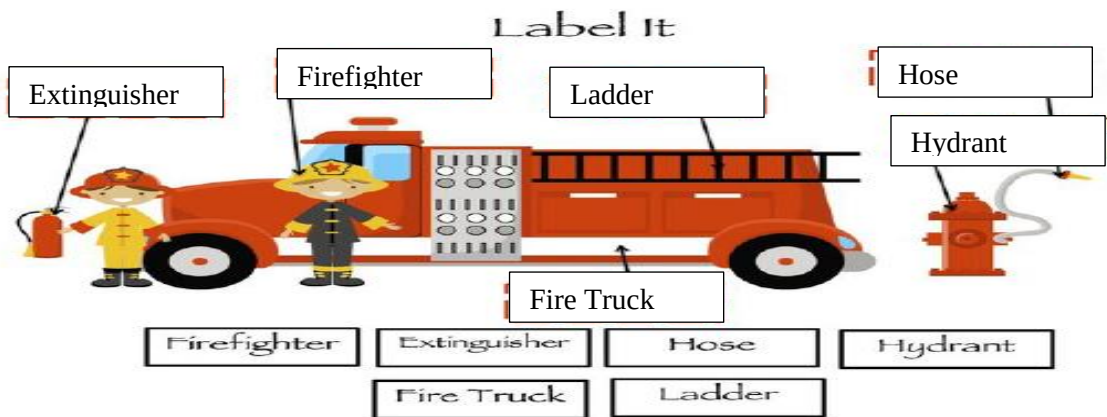
### Expected from Students

- Reflect on their designs, identifying strengths and areas for improvement, and justify their design choices using learned concepts.
- Actively listen to feedback, make revisions, and consider ways to enhance their designs.
- Apply concepts to new scenarios or challenges, collaborating with peers to share ideas and build on each other's designs.
- Extend their understanding by exploring how their design can adapt to different situations, demonstrating creativity and problem-solving skills.

## Focus

# Focus

Fill in the blanks:

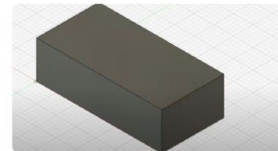


## Practice

<https://www.youtube.com/watch?v=7ZrmNzzKNcM>

## Practice

Create a simple object in fusion360, basic shapes like cubes, spheres, and cylinders.



## Challenge

Use fusion software to design your own fire truck..

